

Final report: customer churn & retention analysis

AUSTRALIAN COMPUTER SOCIETY

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Contents

Executive Summary.....2

1. Summary of Key Findings2

 1.1 Customer Segmentation Analysis (PCA + K-Means)2

 1.2 Churn Prediction Analysis (ANN Model).....3

 1.3 Key Takeaways Table4

2. Identification of Factors Contributing to Churn and Retention.....5

 2.1 Primary Drivers of Customer Churn5

 2.2 Primary Drivers of Customer Retention.....6

 2.3 Synthesis Matrix: Factors Impacting Churn vs. Retention6

3. Recommendations for Targeted Retention Strategies7

 3.1 Segment-Specific Retention Campaigns7

 3.2 Model Integration & Financial ROI Impact9

 3.3 Summary Strategy Matrix..... 10

4. Documentation of Limitations and Proposed Solutions 10

 4.1 Identified Limitations & Technical Challenges 11

 4.2 Proposed Solutions & Mitigation Roadmap 11

 4.3 Summary Matrix: Limitations vs. Proposed Solutions 12

Executive Summary

This report delivers a comprehensive evaluation of customer attrition and retention across **7,043 customer records**. By combining unsupervised customer segmentation (PCA + K-Means) with supervised deep learning (Artificial Neural Networks - ANN), the project identifies key operational risk drivers, profiles high-vulnerability subscriber segments, and delivers a production-ready predictive pipeline. Key findings reveal that churn is concentrated in specific short-term contract segments (~45%–49% churn rates), while the developed ANN model successfully identifies **71.93% of actual churners** prior to cancellation, preserving critical recurring revenue streams.

1. Summary of Key Findings

The objective of this analysis was to address high customer churn rates by identifying key drivers of customer attrition and predicting which subscribers are at risk. The evaluation combined **unsupervised customer segmentation (PCA + K-Means Clustering)** with **supervised deep learning (Artificial Neural Networks - ANN)** across **7,043 customer records**.

1.1 Customer Segmentation Analysis (PCA + K-Means)

To segment the customer base without target bias, Principal Component Analysis (PCA) was used to reduce dimensionality while preserving underlying patterns. Using the **Elbow Method** and **Silhouette Score analysis**, the dataset was segmented into **5 distinct customer clusters**.

Customer Segmentation Summary (PCA + K-Means)						
Cluster	Cluster Size	Churn Rate (%)	Senior Citizens (%)	One-Year Contract (%)	Two-Year Contract (%)	Month-to-month Contract (%) Risk Level
1	1483	49.3	23.6	22.9	33.7	43.4 High
3	1273	45.4	20.2	15.3	13.6	71.1 High
2	1406	20.6	6.0	23.6	23.3	53.1 Medium
0	1769	13.3	21.2	20.5	24.2	55.3 Low
4	1112	3.2	6.7	22.0	24.0	54.0 Very Low

The analysis reveals that customer churn is **heavily concentrated within specific risk groups** rather than uniformly distributed across the business:

- **Cluster 1 (High Risk – Highest Attrition):**
 - **Size:** 1,483 customers
 - **Churn Rate:** 49.3%

- **Key Profile:** Represents the highest-risk group. Driven by a high concentration of senior citizens (**23.6%**) and lower adoption of 1-year contracts (**22.9%**).
- **Cluster 3 (High Risk – Contract Hesitant):**
 - **Size:** 1,273 customers
 - **Churn Rate:** **45.4%**
 - **Key Profile:** High churn risk heavily linked to lack of long-term commitment, exhibiting the lowest 2-year contract adoption rate across all clusters (**13.6%**).
- **Cluster 2 (Medium Risk – Watchlist):**
 - **Size:** 1,406 customers
 - **Churn Rate:** **20.6%**
 - **Key Profile:** Moderate churn risk with low senior citizen representation (**6.0%**) and balanced contract lengths.
- **Cluster 0 (Low Risk – Stable Base):**
 - **Size:** 1,769 customers
 - **Churn Rate:** **13.3%**
 - **Key Profile:** Large, stable segment supported by strong 2-year contract adoption (**24.2%**).
- **Cluster 4 (Very Low Risk – Core Loyalists):**
 - **Size:** 1,112 customers
 - **Churn Rate:** **3.2%**
 - **Key Profile:** The most loyal customer group, characterized by high overall contract commitment (**24.0%** 2-year contracts).

1.2 Churn Prediction Analysis (ANN Model)

A fully connected feed-forward **Artificial Neural Network (ANN)** was developed using TensorFlow/Keras to predict binary customer churn (0 = No Churn, 1 = Churn).

Model Design & Class Imbalance Handling

- **Architecture:** Consists of 12 input features, 3 hidden dense layers (64, 32, and 16 neurons with ReLU activation), 2 Dropout layers (30% and 20%) to prevent overfitting, and a single sigmoid output neuron.
- **Class Weighting:** Because the dataset contains significantly fewer churners (1,869 churn vs. 5,174 no churn), Scikit-learn balanced class weights (**0.6806 for No Churn; 1.8843 for Churn**) were applied during training to prioritise detecting minority churn instances.

Test Evaluation Performance (1,409 Independent Test Records)

- **Accuracy: 74.10%** — Overall proportion of correct predictions across all test instances.
- **ROC-AUC: 80.86%** — Confirms good discriminatory ability between churn and non-churn customers (validated via Results/roc_curve.png).
- **Churn Recall: 71.93%** — The model successfully identified **269 out of 374 actual churners** in the test set (validated via Results/confusion_matrix.png). High recall is critical for ensuring at-risk customers are not missed by retention teams.
- **Churn Precision: 50.85%** — Approximately 51% of customers predicted to churn did, while 49% were false positives (260 non-churners flagged as potential churners).

1.3 Key Takeaways Table

Analytical Pillar	Key Finding	Strategic Business Impact
Segmentation	Clusters 1 and 3 account for most of the churn risk (~45%–49% churn rates).	Retention marketing budgets should be concentrated directly on these two high-risk clusters.
Predictive Power	The ANN captures ~72% of actual churners before they terminate service.	Proactive outreach saves nearly three-quarters of churn-prone recurring revenue streams.
Model	Precision stands at 50.85%	Threshold adjustments are required to

Analytical Pillar	Key Finding	Strategic Business Impact
Precision	(260 false positives).	prevent spending promotional budgets on non-churning customers.

2. Identification of Factors Contributing to Churn and Retention

Understanding the operational, behavioural, and demographic drivers of customer turnover is essential for building targeted retention programs. By synthesising the predictions of the Artificial Neural Network (ANN) model with the profiles generated during PCA + K-Means customer segmentation, several key drivers of both churn and retention were identified.

2.1 Primary Drivers of Customer Churn

1. **Short-Term Contractual Commitments (Month-to-Month Contracts):**

Contract type emerged as the single most critical factor influencing customer churn. Customers on Month-to-Month contracts exhibit significantly higher churn rates compared to those on long-term agreements. In **Cluster 3** (45.4% churn rate), adoption of long-term two-year contracts is the lowest across the entire dataset, at just **13.6%**. Short-term arrangements eliminate switching barriers, making it effortless for customers to churn whenever they experience service disruptions, price hikes, or competitor offers.

2. **Demographic Vulnerability (Senior Citizens):**

Age demographics play a distinct role in subscriber retention patterns. **Cluster 1**, which recorded the highest churn rate across all segments at **49.3%**, contains the highest concentration of senior citizens at **23.6%**. In contrast, segments with low churn rates, such as Cluster 2 (20.6% churn) and Cluster 4 (3.2% churn), have senior citizen representation of only **6.0%** and **6.7%** respectively. Senior customers may be more sensitive to fixed monthly costs, service complexities, or poor digital support experiences.

3. **Service Subscriptions & Financial Factors:**

The underlying dataset features indicate that higher monthly recurring charges and specific service types drive attrition. Customers subscribed to premium services like Fiber Optic

internet without long-term contract discounts face higher MonthlyCharges. These higher charges, combined with month-to-month terms, create high price sensitivity.

2.2 Primary Drivers of Customer Retention

1. Long-Term Contract Adoption (1-Year & 2-Year Agreements):

Contractual duration is the most powerful stabilising factor for the company's subscriber base. **Cluster 4** represents the most stable customer group, with an exceptionally low churn rate of 3.2% and a **24.0%** two-year contract adoption rate. Similarly, **Cluster 0** achieves a low churn rate of **13.3%** while maintaining the highest two-year contract adoption rate across all clusters at **24.2%**.

2. Customer Account Maturity & Service Bundling:

Higher tenure acts as a natural buffer against churn, as customers who remain beyond the initial onboarding phase demonstrate significantly higher lifetime retention. Additionally, features such as having Dependents and subscribing to bundled family plans or multiple lines contribute positively to long-term account stability.

2.3 Synthesis Matrix: Factors Impacting Churn vs. Retention

Factor / Feature	Impact Direction	Primary Segment Affected	Operational Insight
Month-to-Month Contract	Increases Churn Risk	Clusters 1 & 3	High flexibility leads to low loyalty; customers leave easily.
Senior Citizen Status	Increases Churn Risk	Cluster 1 (23.6% senior)	Requires simplified plans and dedicated customer care.
High Monthly Charges	Increases Churn Risk	Premium Fiber Optic Users	High pricing without contract discounts drives bill shock.
2-Year Contract	Increases Retention	Clusters 0 & 4	Strongly locks in commitment and drops churn to as low as

Factor / Feature	Impact Direction	Primary Segment Affected	Operational Insight
			3.2%.
High Tenure	Increases Retention	Core Loyalists	Long-standing accounts exhibit low switching probability.

3. Recommendations for Targeted Retention Strategies

Based on the empirical findings, the company can transition from generic, company-wide offers to **data-driven, targeted interventions** that optimise customer lifetime value (LTV) and marketing ROI.

3.1 Segment-Specific Retention Campaigns

1. High-Risk Cluster 1 (Focus: Senior Citizens & Contract Transition)

- **Target Audience:** Cluster 1 customers (49.3% churn rate, 23.6% senior citizen composition).
- **Strategy:** Address price sensitivity and ease of service for senior demographics while driving contract stabilization.
- **Actionable Tactics:**
 - **Dedicated Customer Support:** Establish a specialized, easily accessible customer care phone line tailored for senior citizens to resolve billing or technical difficulties promptly.
 - **Simplified Senior Value Plans:** Introduce discounted, fixed-rate 12-month basic service bundles that protect seniors against unexpected price fluctuations.
 - **Contract Migration Incentive:** Offer a modest monthly bill credit (e.g., \$5–\$10 off/month) in exchange for transitioning from month-to-month terms to a 1-year agreement.

2. High-Risk Cluster 3 (Focus: Contract Conversion & Value Bundling)

- **Target Audience:** Cluster 3 customers (45.4% churn rate, lowest 2-year contract adoption at 13.6%).
- **Strategy:** Convert contract-hesitant subscribers into multi-year commitments through value enhancement rather than penalties.
- **Actionable Tactics:**
 - **Contract Upgrade Promos:** Deploy targeted digital campaigns offering free premium feature add-ons (e.g., streaming service add-ons or speed boosts) contingent on signing a 1-year or 2-year agreement.
 - **Early Term Loyalty Discounts:** Provide tiered pricing where monthly costs drop progressively after 6 and 12 months of continuous service.

3. Medium-Risk Cluster 2 (Focus: Proactive Engagement & Monitoring)

- **Target Audience:** Cluster 2 customers (20.6% churn rate).
- **Strategy:** Prevent mid-tier risk customers from escalating into high-risk segments.
- **Actionable Tactics:**
 - **Automated Health Checks:** Implement periodic usage and satisfaction surveys (NPS) to catch service friction early.
 - **Cross-Selling Incentives:** Offer discounted bundle options (e.g., adding phone service or multiple lines) to increase product attachment.

4. Low/Very Low-Risk Clusters 0 & 4 (Focus: Advocacy & Reward)

- **Target Audience:** Clusters 0 and 4 customers (13.3% and 3.2% churn rates respectively).
- **Strategy:** Maintain high satisfaction, reward loyalty, and leverage for referral growth.
- **Actionable Tactics:**
 - **VIP Loyalty Program:** Provide surprise rewards (e.g., annual tenure gifts, priority customer service) to reinforce long-term brand loyalty.
 - **Referral Bonus Programs:** Encourage low-risk loyalists to refer friends or family in exchange for bill credits.

3.2 Model Integration & Financial ROI Impact

1. Operationalise Technical Artifacts

- **Deploy Saved Model:** Integrate the saved model artifact (Models/customer_churn_ann_model.keras) directly into the customer relationship management (CRM) database pipeline.
- **Batch Predictions:** Execute automated monthly batch scoring across all active subscriber accounts.

2. Threshold-Based Interventions

To balance retention expenditures against attrition risks, operational workflows should trigger dynamically based on predicted probability ranges:

ANN Predicted Probability	Assigned Risk Level	Triggered Operational Action
$\geq 75\%$ Probability	Critical Churn Risk	Immediate direct contact from retention specialists with high-value contract conversion offers.
50% – 74% Probability	Moderate Risk	Automated digital marketing campaigns featuring contract upgrades and discounted service add-ons.
$< 50\%$ Probability	Low Risk	Standard service, inclusion in routine loyalty and newsletter communications.

3. Financial Impact & Revenue Preservation

- **Quantified ROI:** Capturing 269 actual churners out of 374 in the test set (71.93% recall) translates directly into preserving **~72% of at-risk subscription revenue**.
- Assuming an average monthly charge of **\$65/user**, preventing these cancellations preserves over **\$17,400 in monthly recurring revenue** (\$209,000+ annually) from the test cohort alone, far outweighing the operational cost of targeted retention offers.

3.3 Summary Strategy Matrix

Strategy Area	Recommended Intervention	Target Group	Primary Business Objective
Contract Stabilization	Incentivize migration from Month-to-Month to 1/2-Year plans.	Clusters 1 & 3	Reduce churn rates from ~47% toward company baseline.
Demographic Care	Tailored support channels & simple fixed-rate plans.	Cluster 1 Seniors (23.6%)	Improve customer satisfaction and LTV in vulnerable group.
Predictive Action	Embed saved ANN model (customer_churn_ann_model.keras) in CRM.	Entire Active Base	Capture 72% of potential churners before cancellation.
Loyalty Retention	Tenure rewards and referral incentives.	Clusters 0 & 4	Safeguard core stable revenue base.

4. Documentation of Limitations and Proposed Solutions

Addressing identified technical, data, and operational challenges is essential before deploying the analytical pipeline into production.

4.1 Identified Limitations & Technical Challenges

1. Retention of Uncleaned Duplicate Data Records:

During initial data inspection, **302 duplicate rows** were detected in `customer_churn_preprocessed_Data.csv`. However, the preprocessing scripts reported these duplicates without dropping them prior to model training. This can introduce artificial weighting or data leakage between train/test splits.

2. High False-Positive Rate (Moderate Model Precision):

While the ANN achieved a strong recall of 71.93% (269 out of 374 actual churners captured), its precision was 50.85% (260 false positives in the test set). Targeting non-churning customers with promotional discounts results in unnecessary marketing expenses and margin erosion.

3. Static Decision Thresholding (^{0.50} Classification Cutoff):

The model was trained with a default classification decision threshold of ^{0.50} and fixed network parameters (3 hidden layers: 64, 32, 16 neurons). A default threshold of ^{0.50} assumes equal business costs for false positives and false negatives, which is rarely true in commercial environments.

4. Absence of Baseline Model Benchmarking:

The ANN model was implemented as the sole supervised classifier without direct performance comparisons against standard machine learning algorithms (e.g., Random Forest, XGBoost). Without benchmarking, it is difficult to verify if deep learning provides sufficient performance lift over simpler, more interpretable models.

4.2 Proposed Solutions & Mitigation Roadmap

- Data Cleaning Pipeline Enhancement:** Collaborate with Data Engineering to formalize deduplication rules in the preprocessing pipeline prior to model ingestion, ensuring dataset integrity and eliminating data leakage.
- Decision Threshold Calibration:** Conduct financial cost-benefit optimization to tune the classification threshold away from ^{0.50}. Aligning the cutoff with campaign costs versus customer lifetime value will minimize false positives and maximize campaign ROI.
- Automated Hyperparameter Optimization:** Implement automated hyperparameter tuning (e.g., KerasTuner or Optuna) across layer depth, neuron

counts, dropout rates, and learning rates to elevate the F1-score beyond the baseline of **59.58%**.

4. **Cross-Model Benchmarking:** Train and benchmark baseline algorithms—specifically Logistic Regression, Random Forest, and Gradient Boosted Trees (XGBoost)—against the ANN using identical evaluation splits.

4.3 Summary Matrix: Limitations vs. Proposed Solutions

Limitation / Challenge	Root Cause	Business / Technical Impact	Proposed Solution & Mitigation
Uncleaned Duplicates	302 duplicate rows identified but retained in dataset.	Risk of data leakage and overrepresentation of specific profiles.	Update pre-processing scripts to filter duplicate records prior to model splitting.
Moderate Precision (50.85%)	260 false positive predictions in test set.	Unnecessary marketing costs spent on customers who would not churn.	Calibrate classification threshold based on financial retention costs.
Fixed Architecture	Fixed hidden layers (64-32-16) and default 0.50 threshold.	Model may perform below its optimal predictive capacity.	Execute automated hyperparameter search via KerasTuner.
No Benchmark Models	ANN developed in isolation without comparative evaluation.	Uncertainty regarding whether deep learning is required for this dataset.	Benchmark ANN against Random Forests and Gradient Boosted Trees.